

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**Product Identifier**

**Product Description:** 2,3,5,6-Tetrafluorothiophenol  
**Cat No. :** **B23483**  
**Synonyms** 2,3,5,6-Tetrafluorobenzenethiol  
**CAS No** 769-40-4  
**Molecular Formula** C6 H2 F4 S

**Relevant identified uses of the substance or mixture and uses advised against**

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

**Details of the supplier of the safety data sheet**
**Importer**

Fisher Scientific Korea  
 D5,D6, Incheon Airport Logistics Complex  
 150, Gonghangdong-Ro 296 Beon-Gil  
 Jung-Gu, Incheon  
 Tel: +82-1661-9555  
 Fax: +82-2-2023-0603

**Supplier**

Thermo Fisher Scientific Chemicals, Inc.  
 30 Bond Street  
 Ward Hill, MA 01835-8099

**E-mail address** Chem.KR@thermofisher.com

**Emergency Telephone Number**

Emergency telephone: Medical: +(82) 070-7686-0086 or + 1-703-741-5970  
 CHEMTREC: 080 822 1374 (Local), CHEMTREC: 1-800-424-9300 or + 1-703-527-3887  
 Korea: 00-308-13-2549 (24 hours a day, 7 days a week)

## SECTION 2: HAZARDS IDENTIFICATION

**Classification of the substance or mixture**
**Physical hazards**

Flammable liquids Category 3

**Health hazards**

|                                                    |            |
|----------------------------------------------------|------------|
| Skin Corrosion/Irritation                          | Category 2 |
| Serious Eye Damage/Eye Irritation                  | Category 2 |
| Specific target organ toxicity - (single exposure) | Category 3 |

**Environmental hazards**

Based on available data, the classification criteria are not met

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## Label Elements



### Signal Word

### Warning

#### Hazard Statements

H226 - Flammable liquid and vapor  
H315 - Causes skin irritation  
H319 - Causes serious eye irritation  
H335 - May cause respiratory irritation

#### Precautionary Statements

##### Prevention

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P233 - Keep container tightly closed  
P240 - Ground and bond container and receiving equipment  
P241 - Use explosion-proof electrical/ ventilating/ lighting equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P264 - Wash hands and face thoroughly after handling  
P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P271 - Use only outdoors or in a well-ventilated area

##### Response

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam for extinction  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P321 - Specific treatment (see supplemental first aid instructions on this label)  
P332 + P313 - If skin irritation occurs: Get medical advice/attention  
P362 + P364 - Take off contaminated clothing and wash it before reuse  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P337 + P313 - If eye irritation persists: Get medical advice/attention  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P312 - Call a POISON CENTER or doctor if you feel unwell

##### Storage

P403 + P235 - Store in a well-ventilated place. Keep cool  
P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P405 - Store locked up

##### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

Stench

This product does not contain any known or suspected endocrine disruptors

## NFPA

Health  
2

Flammability  
2

Instability  
0

Physical hazards  
N/A

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

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## 3.1. Substances

| Component                       | Common Name                     | CAS No   | Index No   | Weight % |
|---------------------------------|---------------------------------|----------|------------|----------|
| 2,3,5,6-Tetrafluorobenzenethiol | 2,3,5,6-Tetrafluorobenzenethiol | 769-40-4 | Not listed | 99 - 100 |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

|                                           |                                                                                                                                                  |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                  |
| <b>Skin Contact</b>                       | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get medical attention.                     |
| <b>Ingestion</b>                          | Clean mouth with water. Get medical attention.                                                                                                   |
| <b>Inhalation</b>                         | Remove from exposure, lie down. Remove to fresh air. If not breathing, give artificial respiration. Get medical attention.                       |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |

### Most important symptoms and effects, both acute and delayed

Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

### Indication of any immediate medical attention and special treatment needed

**Notes to Physician** Treat symptomatically. Symptoms may be delayed.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### Suitable Extinguishing Media

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam. Water mist may be used to cool closed containers.

#### Extinguishing media which must not be used for safety reasons

No information available.

### Special hazards arising from the substance or mixture

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Sulfur oxides, Sulfides, Gaseous hydrogen fluoride (HF).

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

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## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental precautions

See Section 12 for additional Ecological Information.

### Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Avoid contact with skin and eyes. Do not breathe mist/vapors/spray. Handle product only in closed system or provide appropriate exhaust ventilation. Use spark-proof tools and explosion-proof equipment. Use only non-sparking tools. Keep away from open flames, hot surfaces and sources of ignition. Take precautionary measures against static discharges.

### Conditions for Safe Storage, Including any Incompatibilities

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep away from heat, sparks and flame. Flammables area. Keep container tightly closed in a dry and well-ventilated place.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component                       | CAS No   | Korea      | ACGIH TLV  | OSHA PEL   |
|---------------------------------|----------|------------|------------|------------|
| 2,3,5,6-Tetrafluorobenzenethiol | 769-40-4 | Not listed | Not listed | Not listed |

| Component                       | CAS No   | European Union | The United Kingdom | Germany    |
|---------------------------------|----------|----------------|--------------------|------------|
| 2,3,5,6-Tetrafluorobenzenethiol | 769-40-4 | Not listed     | Not listed         | Not listed |

### ACGIH - Biological Exposure Indices

| Component                       | CAS No   | ACGIH - Biological Exposure Indices |
|---------------------------------|----------|-------------------------------------|
| 2,3,5,6-Tetrafluorobenzenethiol | 769-40-4 | Not listed                          |

### Exposure Controls

#### Engineering Measures

Use explosion-proof electrical/ventilating/lighting equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

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## Personal protective equipment

|                          |                                                                          |
|--------------------------|--------------------------------------------------------------------------|
| Eye Protection           | Goggles                                                                  |
| Hand Protection          | Protective gloves                                                        |
| Skin and body protection | Wear appropriate protective gloves and clothing to prevent skin exposure |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                               |                                                                                      |
|-------------------------------|--------------------------------------------------------------------------------------|
| Personal protective equipment | Use only those certified by the Korea Occupational Safety and Health Administration. |
| Respiratory Protection        | No protective equipment is needed under normal use conditions                        |

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice

**Environmental exposure controls** No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

**Appearance (Physical State, Color, etc.)** Light yellow Liquid

|                |                          |
|----------------|--------------------------|
| Odor           | Stench                   |
| Odor Threshold | No data available        |
| pH             | No information available |

|                     |                                                           |
|---------------------|-----------------------------------------------------------|
| Melting Point/Range | No data available                                         |
| Softening Point     | No data available                                         |
| Boiling Point/Range | 152 - 153 °C / 305.6 - 307.4 °F @ 760 mmHg                |
| Flash Point         | 48 °C / 118.4 °F <b>Method -</b> No information available |

|                          |                       |
|--------------------------|-----------------------|
| Evaporation Rate         | No data available     |
| Flammability (solid,gas) | Not applicable Liquid |
| Explosion Limits         | No data available     |

|                              |                               |
|------------------------------|-------------------------------|
| Vapor Pressure               | No data available             |
| Vapor Density                | No data available (Air = 1.0) |
| Specific Gravity / Density   | No data available             |
| Bulk Density                 | Not applicable Liquid         |
| Water Solubility             | No information available      |
| Solubility in other solvents | No information available      |

### Partition Coefficient (n-octanol/water)

| Component                       | CAS No   | log Pow           |
|---------------------------------|----------|-------------------|
| 2,3,5,6-Tetrafluorobenzenethiol | 769-40-4 | No data available |

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|                           |                          |                                        |
|---------------------------|--------------------------|----------------------------------------|
| Autoignition Temperature  | No data available        |                                        |
| Decomposition Temperature | No data available        |                                        |
| Viscosity                 | No data available        |                                        |
| Explosive Properties      |                          | explosive air/vapour mixtures possible |
| Oxidizing Properties      | No information available |                                        |

|                   |            |
|-------------------|------------|
| Molecular Formula | C6 H2 F4 S |
| Molecular Weight  | 182.14     |

## SECTION 10: STABILITY AND REACTIVITY

|                   |                                            |
|-------------------|--------------------------------------------|
| <u>Reactivity</u> | None known, based on information available |
|-------------------|--------------------------------------------|

|                           |                                 |
|---------------------------|---------------------------------|
| <u>Chemical Stability</u> | Stable under normal conditions. |
|---------------------------|---------------------------------|

### Possibility of Hazardous Reactions

|                          |                           |
|--------------------------|---------------------------|
| Hazardous Polymerization | No information available. |
| Hazardous Reactions      | No information available. |

|                            |                                                                                                                          |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <u>Conditions to Avoid</u> | Incompatible products. Exposure to moist air or water. Keep away from open flames, hot surfaces and sources of ignition. |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------|

|                               |                                        |
|-------------------------------|----------------------------------------|
| <u>Incompatible Materials</u> | Strong oxidizing agents. Strong bases. |
|-------------------------------|----------------------------------------|

### Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Sulfur oxides. Sulfides. Gaseous hydrogen fluoride (HF).

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

|                     |                                                             |
|---------------------|-------------------------------------------------------------|
| Product Information | No acute toxicity information is available for this product |
|---------------------|-------------------------------------------------------------|

#### Information on expected route of exposure

|            |                                                                                                                  |
|------------|------------------------------------------------------------------------------------------------------------------|
| Inhalation | Avoid breathing vapors or mists.                                                                                 |
| Ingestion  | May be harmful if swallowed.                                                                                     |
| Eyes       | Avoid contact with eyes. Irritating to eyes. Vapor may cause irritation.                                         |
| Skin       | Avoid contact with skin. May cause irritation. Prolonged skin contact may defat the skin and produce dermatitis. |

#### Information on Health Hazards

|                     |                   |
|---------------------|-------------------|
| (a) acute toxicity; | .                 |
| Oral                | No data available |
| Dermal              | No data available |
| Inhalation          | No data available |

| Component | CAS No | LD50 Oral | LD50 Dermal | LC50 Inhalation |
|-----------|--------|-----------|-------------|-----------------|
|-----------|--------|-----------|-------------|-----------------|

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|                                 |          |                   |                   |                   |
|---------------------------------|----------|-------------------|-------------------|-------------------|
| 2,3,5,6-Tetrafluorobenzenethiol | 769-40-4 | No data available | No data available | No data available |
|---------------------------------|----------|-------------------|-------------------|-------------------|

(b) skin corrosion/irritation; Category 2

(c) serious eye damage/irritation; Category 2

(d) respiratory or skin sensitization;  
Respiratory No data available  
Skin No data available

| Component                       | CAS No   | Test method       | Test species      | Study result      |
|---------------------------------|----------|-------------------|-------------------|-------------------|
| 2,3,5,6-Tetrafluorobenzenethiol | 769-40-4 | No data available | No data available | No data available |

(e) germ cell mutagenicity; No data available

| Component                       | CAS No   | Test method       | Test species      | Study result      |
|---------------------------------|----------|-------------------|-------------------|-------------------|
| 2,3,5,6-Tetrafluorobenzenethiol | 769-40-4 | No data available | No data available | No data available |

(f) carcinogenicity; No data available

| Component                       | CAS No   | Test method       | Test species / Duration | Study result      |
|---------------------------------|----------|-------------------|-------------------------|-------------------|
| 2,3,5,6-Tetrafluorobenzenethiol | 769-40-4 | No data available | No data available       | No data available |

There are no known carcinogenic chemicals in this product

| Component                       | CAS No   | IARC       | NTP        | ACGIH      | OSHA       | UK         |
|---------------------------------|----------|------------|------------|------------|------------|------------|
| 2,3,5,6-Tetrafluorobenzenethiol | 769-40-4 | Not listed | Not listed | Not listed | Not listed | Not listed |

(g) reproductive toxicity; No data available

| Component                       | CAS No   | Test method       | Test species / Duration | Study result      |
|---------------------------------|----------|-------------------|-------------------------|-------------------|
| 2,3,5,6-Tetrafluorobenzenethiol | 769-40-4 | No data available | No data available       | No data available |

(h) STOT-single exposure; Category 3  
Results / Target organs Respiratory system.

(i) STOT-repeated exposure; No data available  
Target Organs No information available.

(j) aspiration hazard; No data available

## Other Adverse Effects

Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

| Component                       | CAS No   | EU - Endocrine Disruptors Candidate List | EU - Endocrine Disruptors - Evaluated Substances | Japan - Endocrine Disruptor Information |
|---------------------------------|----------|------------------------------------------|--------------------------------------------------|-----------------------------------------|
| 2,3,5,6-Tetrafluorobenzenethiol | 769-40-4 | Not applicable                           | Not applicable                                   | Not applicable                          |

## SECTION 12: ECOLOGICAL INFORMATION

**Ecotoxicity effects** Contains no substances known to be hazardous to the environment or that are not

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degradable in waste water treatment plants.

| Component                       | CAS No   | Freshwater Fish   | Water Flea        | Freshwater Algae  | Microtox          |
|---------------------------------|----------|-------------------|-------------------|-------------------|-------------------|
| 2,3,5,6-Tetrafluorobenzenethiol | 769-40-4 | No data available | No data available | No data available | No data available |

**Persistence and degradability** No information available

**Bioaccumulative potential** No information available

**Mobility in soil** No information available

**Ozone Depletion Potential**

| Component                       | CAS No   | Ozone Depletion Potential |
|---------------------------------|----------|---------------------------|
| 2,3,5,6-Tetrafluorobenzenethiol | 769-40-4 | Not listed                |

**Other adverse effects** No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

**Waste treatment methods**

**Waste from Residues/Unused Products** Waste is classified as hazardous. Dispose in accordance with the Wastes Control Act (폐기물관리법).

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**Other Information** Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations.

## SECTION 14: TRANSPORT INFORMATION

**Road and Rail Transport**

UN-No UN1993  
Proper Shipping Name FLAMMABLE LIQUID, N.O.S.  
Technical Shipping Name 2,3,5,6-Tetrafluorothiophenol  
Hazard Class 3  
Packing Group III

**IATA**

UN-No UN1993  
Proper Shipping Name FLAMMABLE LIQUID, N.O.S.  
Technical Shipping Name 2,3,5,6-Tetrafluorothiophenol  
Hazard Class 3  
Packing Group III

**IMDG/IMO**

UN-No UN1993  
Proper Shipping Name FLAMMABLE LIQUID, N.O.S.  
Technical Shipping Name 2,3,5,6-Tetrafluorothiophenol  
Hazard Class 3  
Packing Group III  
Marine Pollutant No hazards identified

**Special Precautions for User** No special precautions required

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## SECTION 15: REGULATORY INFORMATION

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

**Legend:** X - Listed '-' - Not Listed

### International Inventories

| Component                       | CAS No   | KECL | TSCA | EINECS    | IECSC | DSL | NDSL | PICCS | ENCS | ISHL | AICS |
|---------------------------------|----------|------|------|-----------|-------|-----|------|-------|------|------|------|
| 2,3,5,6-Tetrafluorobenzenethiol | 769-40-4 | -    | -    | 212-210-1 | -     | -   | -    | -     | -    | -    | -    |

| Component                       | CAS No   | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|---------------------------------|----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| 2,3,5,6-Tetrafluorobenzenethiol | 769-40-4 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

| Component                       | CAS No   | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential |
|---------------------------------|----------|----------------|------------------------------|---------------------------|
| 2,3,5,6-Tetrafluorobenzenethiol | 769-40-4 | Not applicable | Not applicable               | Not applicable            |

### Korean National Regulations

| Component                       | CAS No   | Act on Registration and Evaluation of Chemical Substances (K-REACH) | Authorised Chemicals | Existing Substances Subject to Registration |
|---------------------------------|----------|---------------------------------------------------------------------|----------------------|---------------------------------------------|
| 2,3,5,6-Tetrafluorobenzenethiol | 769-40-4 | Not applicable                                                      | Not applicable       | Not applicable                              |

| Component                       | CAS No   | Chemical Control Act - Toxic Chemicals | Chemical Control Act - Prohibited Chemicals | Chemical Control Act - Use Restricted Chemicals |
|---------------------------------|----------|----------------------------------------|---------------------------------------------|-------------------------------------------------|
| 2,3,5,6-Tetrafluorobenzenethiol | 769-40-4 | Not applicable                         | Not applicable                              | Not applicable                                  |

| Component                       | CAS No   | Chemical Control Act - Accident Precaution Chemicals (% in mixtures) | Chemical Control Act - Accident Precaution Chemicals - Quantity limits Storage (% in mixtures) | Chemical Control Act - Accident Precaution Chemicals - Quantity limits Manufacture/Use (% in mixtures) |
|---------------------------------|----------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| 2,3,5,6-Tetrafluorobenzenethiol | 769-40-4 | Not applicable                                                       | Not applicable                                                                                 | Not applicable                                                                                         |

| Component                       | CAS No   | Waste Control Law | Ministry of Environment - CMR risk | Ministry of Environment - Critically Controlled Substance |
|---------------------------------|----------|-------------------|------------------------------------|-----------------------------------------------------------|
| 2,3,5,6-Tetrafluorobenzenethiol | 769-40-4 | Not applicable    | Not applicable                     | Not applicable                                            |

| Component                       | CAS No   | ISHA - Harmful Agents Subject to Work Environment Monitoring | ISHA - Prohibited substances | ISHA - Substances requiring permission |
|---------------------------------|----------|--------------------------------------------------------------|------------------------------|----------------------------------------|
| 2,3,5,6-Tetrafluorobenzenethiol | 769-40-4 | Not applicable                                               | Not applicable               | Not applicable                         |

| Component                       | CAS No   | ISHA - Substances subject to control | ISHA - Harmful Agents Requiring Health Examination | ISHA - Permissible Exposure Limits |
|---------------------------------|----------|--------------------------------------|----------------------------------------------------|------------------------------------|
| 2,3,5,6-Tetrafluorobenzenethiol | 769-40-4 | Not applicable                       | Not applicable                                     | Not applicable                     |

| Component | CAS No | ISHA - Subject to | ISHA - Threshold Limit | ISHA - Special |
|-----------|--------|-------------------|------------------------|----------------|
|-----------|--------|-------------------|------------------------|----------------|

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|                                 |          | Process Safety Reports<br>(minimum quantity) | Values (TLVs) Chemicals | management materials |
|---------------------------------|----------|----------------------------------------------|-------------------------|----------------------|
| 2,3,5,6-Tetrafluorobenzenethiol | 769-40-4 | Not applicable                               | Not applicable          | Not applicable       |

**National Fire Association - Dangerous Substances** Minimum quantity requiring a permit

| Component                           | CAS No   | Class 1 -<br>Oxidising<br>solids | Class 2 -<br>Flammable<br>solid | Class 3 -<br>Spontaneously<br>Combustible<br>Substances<br>and Dangerous<br>Substances<br>When Wet | Class 4 -<br>Flammable<br>liquids | Class 5 -<br>Self-reactive<br>substances | Class 6 -<br>Oxidising<br>liquids |
|-------------------------------------|----------|----------------------------------|---------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------|------------------------------------------|-----------------------------------|
| 2,3,5,6-Tetrafluorobenz<br>enethiol | 769-40-4 | Not applicable                   | Not applicable                  | Not applicable                                                                                     | Not applicable                    | Not applicable                           | Not applicable                    |

**NFA Dangerous Substances Classification**

Class 4.4 (Group 2 petroleum, insoluble) - 1000 L

**Control Parameters**

| Component                       | CAS No   | Korea      | ACGIH - Biological Exposure Indices |
|---------------------------------|----------|------------|-------------------------------------|
| 2,3,5,6-Tetrafluorobenzenethiol | 769-40-4 | Not listed | Not listed                          |

## US Management Information

**OSHA** - Occupational Safety and Health Administration

Not applicable

| Component                       | CAS No   | Specifically Regulated<br>Chemicals | Highly Hazardous Chemicals |
|---------------------------------|----------|-------------------------------------|----------------------------|
| 2,3,5,6-Tetrafluorobenzenethiol | 769-40-4 | Not applicable                      | Not applicable             |

**CERCLA**

Not applicable

| Component                       | CAS No   | CERCLA Extremely<br>Hazardous Substances<br>RQs | Hazardous Substances<br>RQs | SARA 313 - Threshold<br>Values % |
|---------------------------------|----------|-------------------------------------------------|-----------------------------|----------------------------------|
| 2,3,5,6-Tetrafluorobenzenethiol | 769-40-4 | Not applicable                                  | Not applicable              | Not applicable                   |

**GHS Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567**

Warning.

H226 - Flammable liquid and vapor. H319 - Causes serious eye irritation. H315 - Causes skin irritation. H335 - May cause respiratory irritation.

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

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**LC50** - Lethal Concentration 50%  
**POW** - Partition coefficient Octanol:Water

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road  
**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code  
**OECD** - Organisation for Economic Co-operation and Development  
**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association  
**MARPOL** - International Convention for the Prevention of Pollution from Ships  
**ATE** - Acute Toxicity Estimate  
**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.  
Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.  
First aid for chemical exposure, including the use of eye wash and safety showers.  
Chemical incident response training.

|                         |                                                    |
|-------------------------|----------------------------------------------------|
| <b>Prepared By</b>      | Health, Safety and Environmental Department        |
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## MOEL's Public Notice No. 2023-9 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

### Disclaimer

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**End of Safety Data Sheet**